Reading the Math

A Notation Primer for ISLP — Scalars, Vectors, Hats, and Greeks

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Why This Deck Exists

Math notation is a compressed language

An equation like $\hat{\beta} = (\mathbf{X}^T\mathbf{X})^{-1}\mathbf{X}^T\mathbf{y}$ is one short sentence — *if* you can read it. If you can't, it's a wall.

- ISLP assumes fluency in this language but never teaches it explicitly
- This deck is the **pocket dictionary**: every symbol convention the textbook uses, in one place
- Keep it open next to any chapter — nothing here is more than high-school algebra wearing formal clothes

The Objects: Scalars, Vectors, Matrices

Scalars: single numbers

Written as *italic* letters: n, p, x, y, λ.

Two scalars are so common in ISLP they're practically reserved words:

- n — the **number of observations** (rows of data)
- p — the **number of predictors** (input variables)

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So "$n = 200, p = 3$" in the `Advertising` data means: 200 markets, each described by 3 ad budgets (TV, radio, newspaper).

Vectors: ordered lists of numbers

A vector collects several scalars into one object:

$$x = \begin{pmatrix} x_1 \\ x_2 \\ \vdots \\ x_p \end{pmatrix}$$

- By convention a vector is a **column** (stacked vertically)
- Its **dimension** is how many entries it has — here p
- x_j (one subscript) picks out the j-th entry — a scalar again

When is a vector bold? ISLP's house rule

Most linear algebra texts bold *every* vector ($\mathbf{v}$). ISLP is pickier:

Notation	**Meaning**	**Length**
$\mathbf{y}$, $\mathbf{a}$ (bold lowercase)	a vector with **one entry per observation**	n
x_i, β (plain italic)	any other vector — e.g. one observation's features, or the coefficients	p (or $p + 1$)
$\mathbf{X}$ (bold capital)	a matrix	$n \times p$

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So boldness is a hint about *length*: bold = "as long as the dataset". That's why the response $\mathbf{y}$ is bold but a single observation x_i is not — and why you'll see both in the same equation. Other books bold everything; don't let the font-change throw you when switching sources.

Matrices: grids of numbers

A matrix is written as a bold capital letter, with its size as $\text{rows} \times \text{columns}$:

$$\mathbf{X} = \begin{pmatrix} x_{11} & x_{12} & \cdots & x_{1p} \\ x_{21} & x_{22} & \cdots & x_{2p} \\ \vdots & \vdots & \ddots & \vdots \\ x_{n1} & x_{n2} & \cdots & x_{np} \end{pmatrix} \qquad (n \times p)$$

- x_{ij} — the entry in **row i, column j**: always row first
- A vector is just a matrix with one column ($n \times 1$)
- A scalar is just a 1×1 matrix — it's objects all the way down

The data matrix: one grid, two readings

In ISLP, $\mathbf{X}$ is *the dataset itself* — and you can slice it two ways:

By row

x_i = observation i: one market, one patient, one customer — a vector of its p measurements.

"For observation i…"

By column

$\mathbf{x}_j$ = variable j: all n values of one predictor, e.g. every market's TV budget.

"The predictor X_j…"

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“The predictor X_j…”

When ISLP writes the model $Y = f(X) + \epsilon$, that X means “the predictors” abstractly — not a specific number. Which brings us to…

Capital vs lowercase: random vs realized

- **Capital** X, Y — the **random variable**: the quantity *in general*, before we've measured anything ("income", "whether a customer defaults")
- **Lowercase** x_i, y_i — the **observed value** for observation i: an actual number in your dataset ($x_3 = 41{,}000$)

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That's why the model is stated as $Y = f(X) + \epsilon$ (a claim about the world), but the fitted line is $\hat{y}_i = \hat{\beta}_0 + \hat{\beta}_1 x_i$ (arithmetic on your data).

Check yourself #1

For the `Advertising` data ($n = 200$ markets, $p = 3$ ad channels), classify each symbol — scalar, vector, or matrix? What does it refer to?

$$n \qquad \mathbf{X} \qquad x_i \qquad x_{ij} \qquad Y \qquad y_7$$

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$$n \qquad \mathbf{X} \qquad x_i \qquad x_{ij} \qquad Y \qquad y_7$$

- n — scalar, the number 200
- $\mathbf{X}$ — the 200×3 matrix of all ad budgets
- x_i — vector, the 3 budgets of market i
- x_{ij} — scalar, market i's spend on channel j
- Y — random variable, "sales" in general
- y_7 — scalar, the actual sales of market 7

The Decorations: Sub, Super, Hat

Subscripts: *which one?*

A subscript **indexes** — it answers "which one out of many":

- x_i — the i-th **observation** (rows: $i = 1, \ldots, n$)
- β_j — the coefficient of the j-th **predictor** (columns: $j = 1, \ldots, p$)
- x_{ij} — both at once: observation i, predictor j

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Note the house rule at work: x_i (row i, length p) stays **plain**, while $\mathbf{x}_j$ (column j, length n) is **bold** — same letter, and only the subscript letter plus the boldness tell you which slice of $\mathbf{X}$ you're holding.

Superscripts: usually a power, sometimes a label

- x^2, $\texttt{horsepower}^2$ — a **power**: plain squaring
- $\mathbf{X}^T$ — **transpose**: flip rows and columns (an $n \times p$ matrix becomes $p \times n$)
- $\mathbf{A}^{-1}$ — **matrix inverse**: the matrix analogue of $\frac{1}{a}$; $\mathbf{A}^{-1}\mathbf{A} = \mathbf{I}$ (the identity, matrix-land's "1")
- R^2 — just a name with a square in it (it *is* a squared correlation)

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T and -1 are **operations**, not powers. If a superscript is a letter or sits on a matrix, suspect an operation.

The hat: “our estimate of”

The single most important decoration in this course:

$$\beta_1 \;\; \text{vs} \;\; \hat{\beta}_1 \qquad\qquad f \;\; \text{vs} \;\; \hat{f} \qquad\qquad y_i \;\; \text{vs} \;\; \hat{y}_i$$

- Plain symbol — the **true, unknown** quantity (nature’s secret)
- Hatted symbol — our **estimate** of it, computed from training data

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y_i vs $\hat{y}_i$ deserves care: y_i is what *actually happened*, $\hat{y}_i$ is what our model *predicted*. Their gap $y_i - \hat{y}_i$ is the **residual** — the quantity least squares minimizes.

Bars, tildes, and stars

- $\bar{x}$ — the **sample mean**: $\bar{x} = \frac{1}{n}\sum_{i=1}^{n} x_i$
- $X \sim N(\mu, \sigma^2)$ — the tilde means **"is distributed as"**: X follows a normal distribution with mean μ and variance σ^2
- x^* or x_0 — a **new, unseen** point at which we want a prediction (test point), as opposed to the training points $x_1, \ldots, x_n$

Check yourself #2

Translate each into words:

$$\hat{\beta}_1 \qquad \bar{y} \qquad \mathbf{X}^T \qquad x_{52}^2 \qquad \epsilon \sim N(0, \sigma^2) \qquad \hat{f}(x_0)$$

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- $\hat{\beta}_1$ — our estimate of the first coefficient
- $\bar{y}$ — the average of the observed responses
- $\mathbf{X}^T$ — the data matrix, transposed
- x_{52}^2 — the square of observation 5's value on predictor 2 (context disambiguates 5,2 vs 52!)
- $\epsilon \sim N(0, \sigma^2)$ — the noise is normally distributed, mean 0, variance σ^2
- $\hat{f}(x_0)$ — our fitted model's prediction at the new point x_0

The Greek Cast

Greek letters you'll meet every week

Letter	Name	Typical role in ISLP
β	beta	regression **coefficients** — the stars of Ch. 3–4
ϵ	epsilon	the irreducible **noise** in $Y = f(X) + \epsilon$
μ	mu	a **mean**
σ, σ^2	sigma	standard deviation / **variance**
λ	lambda	**tuning knob** — regularization strength (Ch. 6)
π_k	pi	**prior probability** of class k (Ch. 4 — not 3.14!)
θ	theta	a **generic parameter** when no specific letter applies
ρ	rho	**correlation**
δ_k	delta	**discriminant score** of class k (LDA/QDA)

The unwritten rule: Greek = unknown, Roman = data

- **Roman letters** (x, y, n, p) — things you can *see*: your data and its dimensions
- **Greek letters** (β, μ, σ) — things you *can't* see: the parameters of the process that generated the data
- **Greek with a hat** $(\hat{\beta}, \hat{\mu})$ — your data-based *guess* at the invisible thing

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This is why statisticians write $\hat{\beta}$ rather than b: the notation itself reminds you that **an estimate exists only in relation to an unknown truth**.

One symbol, two jobs: Σ and Π

$$\sum_{i=1}^{n} x_i = x_1 + x_2 + \cdots + x_n \qquad \prod_{i=1}^{n} p_i = p_1 \times p_2 \times \cdots \times p_n$$

Capital sigma = **repeated addition**; capital pi = **repeated multiplication** (you met $\prod$ in the logistic likelihood).

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But in Ch. 4, $\boldsymbol{\Sigma}$ (bold, no limits attached) is the **covariance matrix** of a multivariate normal. Same glyph, different job — **the limits above and below are the giveaway**: with limits it's a sum, without them it's a matrix.

Check yourself #3

From Week 2's slides — name each Greek letter and its role:

$$p(X) = \frac{e^{\beta_0+\beta_1 X}}{1 + e^{\beta_0+\beta_1 X}} \qquad \Pr(Y = k \mid X = x) = \frac{\pi_k f_k(x)}{\sum_{l=1}^{K} \pi_l f_l(x)}$$

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- β_0, β_1 — betas: intercept and slope on the *log-odds* scale
- π_k — pi: the prior probability of class k
- $\sum_{l=1}^{K}$ — capital sigma *with limits*: sum over all K classes
- Bonus: e is Roman — Euler's number 2.718…, a known constant, not a parameter

Functions and Parameters

Function notation: f vs $f(x)$

A function is a **machine**: feed in an input, get an output.

- f — the machine itself (the relationship, the “true pattern”)
- $f(x)$ — the machine’s **output** at a particular input x — a number
- $\hat{f}$ — our estimated copy of the machine; $\hat{f}(x_0)$ — its prediction at a new point

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"Estimating f" — the phrase defining this whole course — means building a machine whose outputs mimic nature's, **without ever seeing nature's blueprints**.

Arguments vs parameters

Two kinds of things live inside a function's parentheses:

- **Arguments** — the inputs that vary: the X in $f(X)$
- **Parameters** — the settings that define *which* function you have: the β's in $\beta_0 + \beta_1 X$

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$p(X)$ in logistic regression is read "the probability, *as a function of* X" — its shape is controlled by parameters β_0, β_1 hiding inside. Fitting = **using data to choose parameter values**; predicting = plugging arguments into the fitted function.

Operators: $E[\cdot], \mathrm{Var}(\cdot), \Pr(\cdot)$

These look like functions, and they are — their input is a *random quantity*:

- $E[Y]$ — the **expected value** (long-run average) of Y
- $\mathrm{Var}(\hat{f}(x_0))$ — the **variance**: how much this quantity would wobble across different training sets
- $\Pr(Y = k \mid X = x)$ – a **conditional probability**: the chance Y is class k, **given that** we know $X = x$ – the "$|$" is read "given"

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None of these are multiplications: $E[Y]$ is not "E times Y". Brackets after an operator name mean "applied to".

Compact shorthands worth memorizing

- $I(\text{condition})$ — the **indicator**: equals 1 if the condition holds, 0 if not. Week 1's error rate $\frac{1}{n}\sum_i I(y_i \neq \hat{y}_i)$ literally says: *count the mistakes, divide by* n
- $\arg\min_\beta \; g(\beta)$ — "the value of β that makes g smallest" ($\arg\max$: largest). Least squares in one line: $\hat{\beta} = \arg\min_\beta \mathrm{RSS}(\beta)$
- $\log$ — in ISLP (and most of ML), always the **natural log** (base e)
- $\mathbb{R}^p$ — the set of all vectors with p real entries; "$x \in \mathbb{R}^p$" = "x is a p-dimensional vector"

Worked Example: Simple Linear Regression, Three Ways

The model, symbol by symbol

$$y_i = \beta_0 + \beta_1 x_i + \epsilon_i, \qquad i = 1, \ldots, n$$

- y_i — observed response of observation i (Roman, lowercase: data)
- x_i — observed predictor of observation i (data)
- β_0, β_1 — unknown intercept and slope (Greek: parameters)
- ϵ_i — observation i's unobservable noise (Greek: unknown)
- The trailing "$i = 1, \ldots, n$" means: **this equation holds n times, once per observation** — it's n equations wearing one line

A dataset small enough to see everything

Three observations ($n = 3, p = 1$) — think "TV spend vs sales":

i	x_i	y_i
1	1	2
2	2	4
3	3	3

The sample means (bars!): $\bar{x} = \frac{1+2+3}{3} = 2, \quad \bar{y} = \frac{2+4+3}{3} = 3.$

Way 1: scalar arithmetic

The Ch. 3 least squares formulas, then plug in:

$$\hat{\beta}_1 = \frac{\sum_{i=1}^{n}(x_i - \bar{x})(y_i - \bar{y})}{\sum_{i=1}^{n}(x_i - \bar{x})^2} = \frac{(-1)(-1) + (0)(1) + (1)(0)}{(-1)^2 + 0^2 + 1^2} = \frac{1}{2}$$

$$\hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x} = 3 - \tfrac{1}{2} \cdot 2 = 2$$

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$$\hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x} = 3 - \tfrac{1}{2} \cdot 2 = 2$$

Fitted line: $\hat{y} = 2 + 0.5x$. Notice the notation working: sums run over i (observations), bars are means, hats appear exactly when data replaces truth.

Way 2: the same data as vectors and a matrix

Stack the n scalar equations into one vector equation $\mathbf{y} = \mathbf{X}\beta + \epsilon$:

$$\underbrace{\begin{pmatrix} 2 \\ 4 \\ 3 \end{pmatrix}}_{\mathbf{y}\ (3\times1)} = \underbrace{\begin{pmatrix} 1 & 1 \\ 1 & 2 \\ 1 & 3 \end{pmatrix}}_{\mathbf{X}\ (3\times2)} \underbrace{\begin{pmatrix} \beta_0 \\ \beta_1 \end{pmatrix}}_{\beta\ (2\times1)} + \underbrace{\begin{pmatrix} \epsilon_1 \\ \epsilon_2 \\ \epsilon_3 \end{pmatrix}}_{\epsilon\ (3\times1)}$$

- The column of 1's is the intercept's seat — it multiplies β_0
- $\mathbf{X}$ here is called the **design matrix**: one row per observation, one column per parameter

Multiplying it out: the two notations are the same thing

Matrix × vector = take each **row** of $\mathbf{X}$, multiply entrywise with β, add up:

$$\mathbf{X}\beta = \begin{pmatrix} 1 \cdot \beta_0 + 1 \cdot \beta_1 \\ 1 \cdot \beta_0 + 2 \cdot \beta_1 \\ 1 \cdot \beta_0 + 3 \cdot \beta_1 \end{pmatrix} = \begin{pmatrix} \beta_0 + \beta_1 \cdot 1 \\ \beta_0 + \beta_1 \cdot 2 \\ \beta_0 + \beta_1 \cdot 3 \end{pmatrix}$$

Row i is exactly the scalar equation $\beta_0 + \beta_1 x_i$. **The matrix form isn't new math — it's the same three equations, holding hands.**

Way 3: the one-line solution

$$\hat{\beta} = (\mathbf{X}^T\mathbf{X})^{-1}\mathbf{X}^T\mathbf{y}$$

Reading right to left, with our numbers:

$$\mathbf{X}^T\mathbf{X} = \begin{pmatrix} 3 & 6 \\ 6 & 14 \end{pmatrix} \qquad \mathbf{X}^T\mathbf{y} = \begin{pmatrix} 9 \\ 19 \end{pmatrix}$$

$$\hat{\beta} = \begin{pmatrix} 3 & 6 \\ 6 & 14 \end{pmatrix}^{-1} \begin{pmatrix} 9 \\ 19 \end{pmatrix} = \frac{1}{6} \begin{pmatrix} 14 & -6 \\ -6 & 3 \end{pmatrix} \begin{pmatrix} 9 \\ 19 \end{pmatrix} = \begin{pmatrix} 2 \\ 0.5 \end{pmatrix}$$

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Same $\hat{\beta}_0 = 2, \hat{\beta}_1 = 0.5$ as the scalar route — and this is **literally what `statsmodels` computes** when you call `.fit()`, for any number of predictors.

Predictions and residuals, both ways

$$\hat{\mathbf{y}} = \mathbf{X}\hat{\beta} = \begin{pmatrix} 2.5 \\ 3 \\ 3.5 \end{pmatrix} \qquad \mathbf{e} = \mathbf{y} - \hat{\mathbf{y}} = \begin{pmatrix} -0.5 \\ 1 \\ -0.5 \end{pmatrix}$$

$$\mathrm{RSS} = \sum_{i=1}^{3} e_i^2 = 0.25 + 1 + 0.25 = 1.5 \qquad \text{or} \qquad \mathrm{RSS} = \mathbf{e}^T\mathbf{e} = 1.5$$

$\mathbf{e}^T\mathbf{e}$ — a row vector times a column vector — *is* the sum of squares. Every Σ-formula in ISLP has a compact matrix twin.

Check yourself #4

Dimension bookkeeping — if $\mathbf{X}$ is $n \times p$ and $\mathbf{y}$ is $n \times 1$, what are the shapes of:

$$\mathbf{X}^T \qquad \mathbf{X}^T\mathbf{X} \qquad \mathbf{X}^T\mathbf{y} \qquad (\mathbf{X}^T\mathbf{X})^{-1}\mathbf{X}^T\mathbf{y}$$

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- $\mathbf{X}^T$: $p \times n$
- $\mathbf{X}^T\mathbf{X}$: $(p \times n)(n \times p) = p \times p$ — inner n's cancel
- $\mathbf{X}^T\mathbf{y}$: $(p \times n)(n \times 1) = p \times 1$
- $(\mathbf{X}^T\mathbf{X})^{-1}\mathbf{X}^T\mathbf{y}$: $(p \times p)(p \times 1) = p \times 1$ — one coefficient per predictor. **Dimensions always chain: inner sizes must match, outer sizes survive.**

Wrap-Up

The cheat sheet

You see	You read
n, p	# observations, # predictors
$\mathbf{y}$ vs x_i, β	bold = length-n vector; plain = other vectors (ISLP rule)
x_{ij}	row i (observation), column j (variable)
$\hat{\theta}$	estimate of the unknown θ
$\bar{x}$	sample mean of x
$\mathbf{X}^T, \mathbf{A}^{-1}$	transpose; matrix inverse